

DS 4400: Machine Learning and Data Mining I
Spring 2022

Project Title: Predicting Rainfall in Australia

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Link to code: <https://github.com/abbycarr/DS4400-ML-Rain>

Powerpoint:

<https://docs.google.com/presentation/d/1Q15BmKupxqBBh27iIRI4oXo0WSaJrtp8DPIqwkOEFRk/edit?usp=sharing>

Link to video recording:

<https://northeastern.zoom.us/rec/share/O1pf0NEvulQGBn-91I3gw34SI8FXviBGHQgRqZbbGMvDefAWE5C5jzBb0gEooNU.erlxd2YB9ks42oz2>

Passcode: +x\$#M4eb

I. Problem Description

For our project, we were trying to implement several ML models that will predict whether or not it will rain a given day, due to various features describing the weather the day before. The problem can be described as a binary classification problem with the two classes being that “it will rain”/ “it will not rain”. Given that weather can greatly affect one’s plans for the day, such as whether they will spend the day outside, what they will wear, etc, this can prove to be helpful in planning purposes for many. Additionally, with the topic of climate change being one of great importance, tracking changes in rainfall prediction could prove to be an investigation of great value.

Before starting our project, we explored various other projects, papers and articles on the topic. One such article focuses on climate trends in Australia[1]. We wanted to gain more insight into whether rainfall patterns are changing over time. This would be interesting as that could mean that the model, based on date, would be more or less likely to predict rainfall. This source talked about the rising temperature in Australia overtime during daytime and nighttime. We also see that region plays a big role in rainfall prediction, as rainfall has increased in most of Northern Australia in the past few decades, vs. a decrease in rainfall in the southwest and southeast. This indicates the importance of accounting for regions in our rainfall prediction model.

The second article we looked at was published by another group using ML in order to predict rainfall in Iowa[2]. For our project, we will be implementing 4 different algorithms, and this group also used different algorithms, so we wanted to gain insight into which of the algorithms appeared to be most promising in terms of accurate and precise prediction. Based on their results, it appeared that the Neural Network, multilayer perceptron model provided the best results as well as SVM, and are models we should therefore consider for implementation.

An interesting find was this review of different Machine Learning models on the general task of predicting rain[3]. As the paper remarks, predicting rain is an important and often-pursued technique, and looking at the different common methods has merit; even more for our purpose, looking at what has worked gives us a good baseline for our expectations working on the Australia rain model but also gives us context for a large scope of workings in the rain modeling field. Some interesting findings were the effects of differences in the time period rain data was taken from.

And finally, a carbon copy of the type of task we wish to perform, this paper is best when seen as a comparison of where we might have gone[4]. It provides very useful information on what type of data cleaning we may have opted for instead as well as how it impacted similar

results between their work and ours. In addition, this paper provides insight into the types of models and motivations behind choosing them.

II. References

1. <https://www.climatechangeinaustralia.gov.au/en/changing-climate/climate-trends/australian-trends/>
2. <https://ieeexplore.ieee.org/document/6297455>
3. https://www.researchgate.net/profile/Aakash-Parmar-2/publication/319503839_Machine_Learning_Techniques_For_Rainfall_Prediction_A_Review/links/59afb922458515150e4c2e4/Machine-Learning-Techniques-For-Rainfall-Prediction-A-Review.pdf
4. <https://arxiv.org/abs/1910.13827>
5. <https://www.kaggle.com/jsphyg/weather-dataset-rattle-package>

III. Dataset and Exploratory Data Analysis

a. Dataset

The dataset we used came from Kaggle [5]. This dataset focuses on various regions in Australia and features corresponding to their weather, drawn from weather stations on a daily basis. Before any data cleaning, there were 145k records comprising 22 feature columns. 4 of the columns are categorical, 1 is Boolean, and the rest are numerical. The boolean column was the target column “RainTomorrow” where the classes were listed as “Yes” or “No”. The following table contains a further breakdown of each of the columns.

Column Name	Description	Data Type
Date	The date of observation	DateTime
Location	The common name of the location of the weather station	String, Categorical
MinTemp	The minimum temperature in degrees celsius	Float (Celcius)
MaxTemp	The maximum temperature in degrees celsius	Float (Celcius)
Rainfall	The amount of rainfall recorded for the day in mm	Float (mm)
Evaporation	The so-called Class A pan evaporation (mm) in the 24 hours to 9am	Float (mm)
Sunshine	The number of hours of bright sunshine in the day.	Float (hours)
WindGustDir	The direction of the strongest wind gust in the 24 hours to midnight	String, Categorical One of many cardinal directions
WindGustSpeed	The speed (km/h) of the strongest	Float (km/hr)

Column Name	Description	Data Type
Date	The date of observation	DateTime
	wind gust in the 24 hours to midnight	
WindDir9am	Direction of the wind at 9am	String, Categorical One of many cardinal directions
WindDir3pm	Direction of the wind at 3pm	String, Categorical One of many cardinal directions
WindSpeed9am	Wind speed (km/hr) averaged over 10 minutes prior to 9am	Float (km/hr)
WindSpeed3pm	Wind speed (km/hr) averaged over 10 minutes prior to 3pm	Float (km/hr)
Humidity9am	Humidity (percent) at 9am	Float (percentage)
Humidity3pm	Humidity (percent) at 3pm	Float (percentage)
Pressure9am	Atmospheric pressure (hpa) reduced to mean sea level at 9am	Float (hpa)
Pressure3pm	Atmospheric pressure (hpa) reduced to mean sea level at 3pm	Float (hpa)
Cloud9am	Fraction of sky obscured by cloud at 9am. This is measured in "oktas", which are a unit of eighths. It records how many	Integer, (oktas) 0 to 8 inclusive
Cloud3pm	Fraction of sky obscured by cloud (in "oktas": eighths) at 3pm. See Cloud9am for a description of the values	Integer, (oktas) 0 to 8 inclusive
Temp9am	Temperature (degrees C) at 9am	Float (Celcius)
Temp3pm	Temperature (degrees C) at 3pm	Float (Celcius)
RainToday	Boolean: 1 if precipitation (mm) in the 24 hours to 9am exceeds 1mm, otherwise 0	Boolean
RainTomorrow	The amount of next day rain in mm. Used to create response variable RainTomorrow. A kind of measure of the "risk".	Float (mm)

b. Exploratory Data Analysis

The first step in our analysis was looking at the quality of our data in terms of completeness and usability of each data column. In terms of columns that would give up the most trouble, we saw that less than $\frac{1}{3}$ of data was available for each of the Evaporation, Sunshine, Rainfall, and Cloud columns which we then removed. We removed the Location column due to the large variation of locations that would add 44 new feature columns if encoded. The remaining rows with NaN values (30,045) were removed.

We then looked at the correlation between features in order to support our feature selection efforts. We first printed out a table with the linear correlation between features, and then the corresponding heat map.

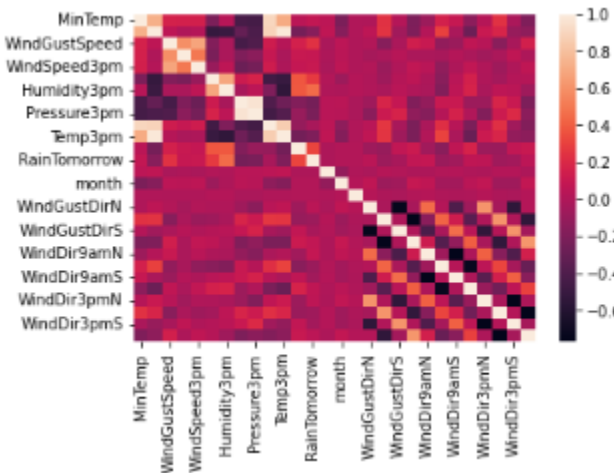
	MinTemp	MaxTemp	WindGustSpeed	WindSpeed9am	WindSpeed3pm
MinTemp	1.000000	0.728610	0.159282	0.150052	0.145654
MaxTemp	0.728610	1.000000	0.056850	-0.005989	0.013626
WindGustSpeed	0.159282	0.056850	1.000000	0.590132	0.678569
WindSpeed9am	0.150052	-0.005989	0.590132	1.000000	0.492155
WindSpeed3pm	0.145654	0.013626	0.678569	0.492155	1.000000
Humidity9am	-0.217067	-0.515588	-0.183934	-0.222205	-0.093797
Humidity3pm	0.026907	-0.497169	-0.021263	-0.018744	0.052981
Pressure9am	-0.433600	-0.312804	-0.446807	-0.202463	-0.285172
Pressure3pm	-0.440856	-0.411641	-0.400743	-0.151562	-0.245085
Temp9am	0.895082	0.894496	0.124632	0.089087	0.126241
Temp3pm	0.703939	0.984276	0.018299	-0.019171	-0.011790
RainToday	0.043948	-0.240498	0.152262	0.097250	0.086494
RainTomorrow	0.077364	-0.167040	0.234628	0.091620	0.095690
year	0.048528	0.062946	-0.027563	-0.014091	-0.027539
month	-0.211504	-0.171276	-0.051801	-0.042637	0.057476
day	0.001575	-0.001384	-0.008812	-0.006671	-0.005200
WindGustDirN	-0.006184	0.058115	0.003732	-0.039966	0.004605
WindGustDirE	0.236121	0.239784	-0.167436	-0.022673	-0.092648
WindGustDirS	0.011241	-0.064723	-0.021431	0.028567	0.001690
WindGustDirW	-0.208069	-0.202330	0.140730	-0.021348	0.071252
WindDir9amN	-0.002341	0.017123	0.004625	-0.021369	0.034362
WindDir9amE	0.150988	0.294265	-0.125451	-0.032322	-0.133435
WindDir9amS	0.090011	-0.025456	-0.006797	0.010572	-0.011101
WindDir9amW	-0.145901	-0.283775	0.062170	-0.020491	0.069909
WindDir3pmN	0.020486	0.101639	0.024387	-0.064941	-0.001806
WindDir3pmE	0.232103	0.175630	-0.157909	-0.023885	-0.074640
WindDir3pmS	-0.019085	-0.102426	-0.033484	0.063355	-0.004145
WindDir3pmW	-0.197892	-0.138989	0.144229	-0.014892	0.064907

	Humidity9am	Humidity3pm	Pressure9am	Pressure3pm	Temp9am
MinTemp	-0.217067	0.026907	-0.433600	-0.440856	0.895082
MaxTemp	-0.515588	-0.497169	-0.312804	-0.411641	0.894496
WindGustSpeed	-0.183934	-0.021263	-0.446807	-0.400743	0.124632
WindSpeed9am	-0.222205	-0.018744	-0.202463	-0.151562	0.089087
WindSpeed3pm	-0.093797	0.052981	-0.285172	-0.245085	0.126241
Humidity9am	1.000000	0.683476	0.116009	0.168641	-0.463127
Humidity3pm	1.000000	1.000000	0.034903	0.040907	-0.201721
Pressure9am	0.116009	-0.034903	1.000000	0.960230	-0.400322
Pressure3pm	0.168641	0.040907	0.960230	1.000000	-0.453182
Temp9am	-0.463127	-0.201721	-0.400322	-0.453182	1.000000
Temp3pm	-0.508436	-0.547082	-0.270688	-0.378721	0.859775
RainToday	0.374266	0.382146	-0.186420	-0.103529	-0.112551
RainTomorrow	0.273124	0.448708	-0.239226	-0.219729	-0.035530
year	0.006598	-0.008687	0.038101	0.034464	0.048055
month	-0.002991	-0.015133	0.039810	0.029779	-0.141460
day	0.015511	0.013623	-0.022651	-0.022531	-0.008015
WindGustDirN	-0.006430	-0.001668	-0.009196	-0.152853	0.032510
WindGustDirE	-0.074124	-0.028515	0.171744	0.111658	0.235887
WindGustDirS	0.020966	0.032729	0.058080	0.164242	-0.031809
WindGustDirW	0.052706	0.003379	-0.196421	-0.133684	-0.216874
WindDir9amN	-0.016518	-0.020678	-0.115900	-0.170653	-0.023626
WindDir9amE	-0.161222	-0.187949	0.098982	0.023264	0.220149
WindDir9amS	0.035753	0.053032	0.106536	0.160200	0.023591
WindDir9amW	0.152589	0.169383	-0.009357	-0.000609	-0.213503
WindDir3pmN	-0.007751	-0.029921	-0.002790	-0.187342	0.060700
WindDir3pmE	-0.048454	0.052088	0.172866	0.118540	0.231890
WindDir3pmS	0.012893	0.044492	0.105549	0.201575	-0.063271
WindDir3pmW	0.023539	-0.071357	-0.210663	-0.151664	-0.185623

	WindDir3pmS	WindDir3pmW
MinTemp	-0.019085	-0.197892
MaxTemp	-0.102426	-0.138989
WindGustSpeed	-0.033484	0.144229
WindSpeed9am	0.063355	-0.014892
WindSpeed3pm	-0.004145	0.064907
Humidity9am	0.012893	0.023539
Humidity3pm	0.044492	-0.071357
Pressure9am	0.105549	-0.210663
Pressure3pm	0.201575	-0.151664
Temp9am	-0.063271	-0.185623
Temp3pm	-0.115212	-0.145150
RainToday	0.069093	0.096182
RainTomorrow	0.055316	0.051963
year	0.014538	-0.012617
month	-0.067814	0.072953
day	-0.017222	0.012409
WindGustDirN	-0.556922	0.014891
WindGustDirE	-0.002444	-0.548185
WindGustDirS	0.607608	-0.052693
WindGustDirW	-0.006740	0.615058
WindDir9amN	-0.402742	0.148327
WindDir9amE	-0.040618	-0.294299
WindDir9amS	0.410348	-0.156081
WindDir9amW	0.058708	0.298521
WindDir3pmN	-0.769317	-0.011010
WindDir3pmE	0.011141	-0.759982
WindDir3pmS	1.000000	-0.044114
WindDir3pmW	-0.044114	1.000000

	WindGustDirS	WindGustDirW	WindDir9amW	WindDir9amE
MinTemp	0.011241	-0.208689	-0.082341	0.150988
MaxTemp	-0.064723	-0.203830	0.017123	0.294265
WindGustSpeed	-0.021431	0.140730	0.094629	-0.125451
WindSpeed9am	0.020567	-0.021348	0.021269	-0.033222
WindSpeed3pm	0.001598	0.071252	0.034362	-0.133435
Humidity9am	0.020966	0.052706	-0.016518	-0.161222
Humidity3pm	0.003379	-0.020678	-0.020678	-0.187949
Pressure9am	0.085080	-0.196421	-0.115900	0.098982
Pressure3pm	0.164242	-0.133684	-0.170653	0.023264
Temp9am	-0.031809	-0.216874	-0.023626	0.220149
Temp3pm	-0.070591	-0.214987	0.008123	0.305687
RainToday	0.067237	0.110280	0.059980	-0.159525
RainTomorrow	-0.037078	0.065986	0.081530	-0.089046
year	0.009671	-0.011900	-0.016736	0.009713
month	-0.062899	0.087075	0.060726	-0.053155
day	-0.017335	0.009750	0.017568	-0.004834
WindGustDirN	-0.748244	-0.036471	0.452190	0.015483
WindGustDirE	-0.010564	-0.759035	-0.151464	0.425403
WindGustDirS	1.000000	-0.010018	-0.423405	-0.028496
WindGustDirW	-0.010018	1.000000	0.117246	-0.381457
WindDir9amN	-0.423405	0.117246	1.000000	-0.068654
WindDir9amE	-0.028496	-0.381457	-0.068654	1.000000
WindDir9amS	-0.459169	-0.125731	-0.771442	0.017492
WindDir9amW	-0.035083	0.401740	-0.013595	-0.732689
WindDir3pmN	-0.547171	-0.036130	0.406567	0.047873
WindDir3pmE	-0.023449	-0.542364	-0.162731	0.311634
WindDir3pmS	0.607608	-0.006740	-0.402742	-0.040618
WindDir3pmW	-0.052693	0.615058	0.148327	-0.294299

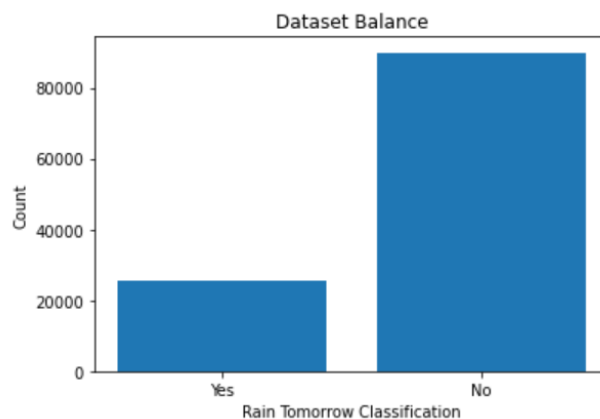
	WindDir9amS	WindDir9amW	WindDir3pmN	WindDir3pmE
MinTemp	0.090011	-0.145901	0.020486	0.232103
MaxTemp	-0.025456	-0.283775	0.101639	0.175630
WindGustSpeed	-0.006797	0.052170	0.024387	-0.157909
WindSpeed9am	0.010572	-0.020491	-0.064041	-0.023885
WindSpeed3pm	-0.011101	0.009909	-0.001806	-0.074640
Humidity9am	0.035753	0.152589	-0.007751	-0.048454
Humidity3pm	0.053032	0.169383	-0.029921	0.052088
Pressure9am	0.106636	-0.009357	-0.002790	0.172866
Pressure3pm	0.160200	0.000609	-0.187342	0.118540
Temp9am	0.029591	-0.213503	0.060700	0.231890
Temp3pm	-0.020113	-0.290971	0.113314	0.181632
RainToday	0.063898	0.163406	-0.009053	-0.089936
RainTomorrow	-0.064682	0.062425	0.058505	-0.067198
year	0.010905	-0.005258	-0.016454	0.010626
month	-0.056855	0.042558	0.066545	-0.066389
day	-0.016717	-0.003844	0.019500	-0.014602
WindGustDirN	-0.410688	-0.062556	0.009098	-0.040537
WindGustDirE	-0.123463	-0.357073	0.009038	0.060295
WindGustDirS	0.459169	0.035083	-0.547171	0.023449
WindGustDirW	-0.125731	0.401740	-0.036130	-0.542364
WindDir9amN	-0.771442	-0.013595	0.406567	-0.162731
WindDir9amE	0.017492	-0.732689	0.047873	0.311634
WindDir9amS	1.000000	-0.016196	-0.369857	0.142363
WindDir9amW	-0.016196	1.000000	-0.096715	-0.259816
WindDir3pmN	-0.369857	-0.096715	1.000000	0.028212
WindDir3pmE	0.142363	-0.259816	0.028212	1.000000
WindDir3pmS	0.410348	0.058708	-0.769317	0.011141
WindDir3pmW	-0.156081	0.298521	-0.011010	-0.759982



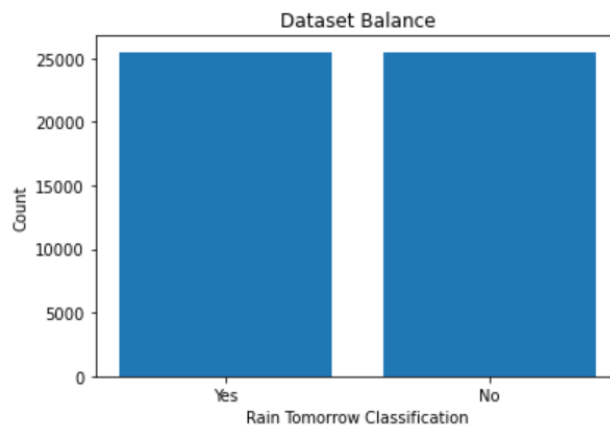
Given that we saw columns with high levels of correlation, we decided to drop one of each pair of columns where the correlation was greater than 0.85.

Before starting any analysis, we first wanted to see the balance of the data between the two classes. We wanted to make sure that there was balance present in our data before implementing any algorithms. As we can see below, there was a great imbalance leaning towards a classification of “No”, therefore we decided to implement undersampling to make the distribution more even.

Before Manipulation:



After Undersampling:



Finally, we used Standard Scalar to normalize the data before moving forward with running the models on the data. There were large discrepancies between the scale of data in the different features included for our dataset, so we wanted to scale in order to provide some normalization.

III. Approach and Methodology

As mentioned above, we took steps to balance the data using undersampling and normalization. For feature selection, we chose the features of most value based on the results of linear correlation between features, and dropping columns with more than $\frac{1}{3}$ of values being N/A.

Some data manipulation was also necessary. For the “RainToday” and “RainTomorrow” columns, where the values were either “Yes” or “No”, we performed one-hot encoding so that the “yes” would now be 1 and the “no” 0.

RainToday	RainTomorrow	RainToday	RainTomorrow
No	No	0	0
No	No	0	0
No	No	0	0
No	No	0	0
No	No	0	0

For the date column, we manipulated the original date column which had everything in the format yyyy-mm-dd and created three new columns, one for the month, day, and year.

Date	year	month	day
2008-12-01	2008	12	1
2008-12-02	2008	12	2
2008-12-03	2008	12	3
2008-12-04	2008	12	4
2008-12-05	2008	12	5

And finally, for all of the columns that implemented the cardinal directions, we implemented one hot encoding so that if one of the N, S, E, or W was included in the original column, the encoded column would contain a 1.

WindDir9am	WindDir9amN	WindDir9amE	WindDir9amS	WindDir9amW
W	0	0	0	1
NNW	1	0	0	1
W	0	0	0	1
SE	0	1	1	0
ENE	1	1	0	0

After all of our data cleaning, feature selection, and undersampling, we were left with 51,012 records and 22 feature columns.

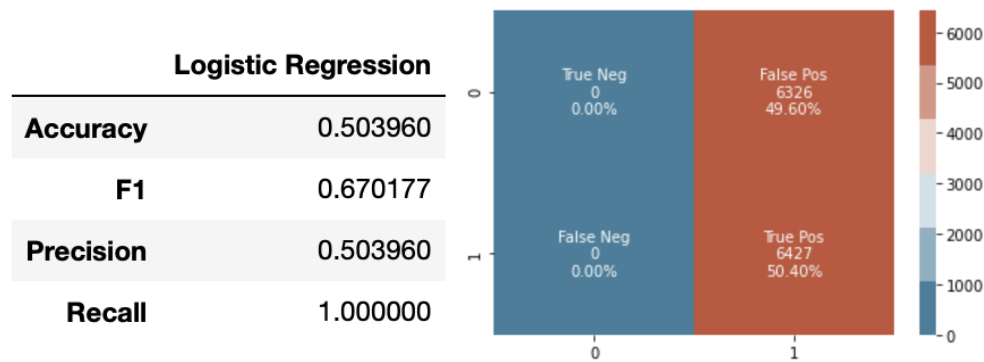
We implemented the following models in our project: Logistic Regression, Naive Bayes, SVM, and Random Forest. These models were implemented in Python and we used modules such as Pandas, Numpy, and Sklearn to create the models, and seaborn and Matplotlib to visualize the results. In order to test our models, we did a random split of the data with 75% going towards the training set, and 25% left for testing.

In order to evaluate our models, we will be using accuracy, recall, precision, and f1 score. We will focus on recall and precision over accuracy as this will provide better insight into whether the model consistently predicts either that it will rain or will not, rather than truly predicting based on the given features.

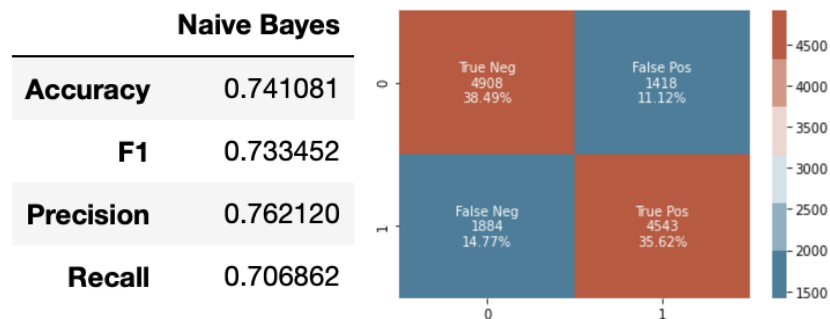
IV. Discussion and Result Interpretation

When looking at each of the four models we used, we can see varying levels of performance based on the four metrics we looked at - accuracy, precision, recall, and F1 score.

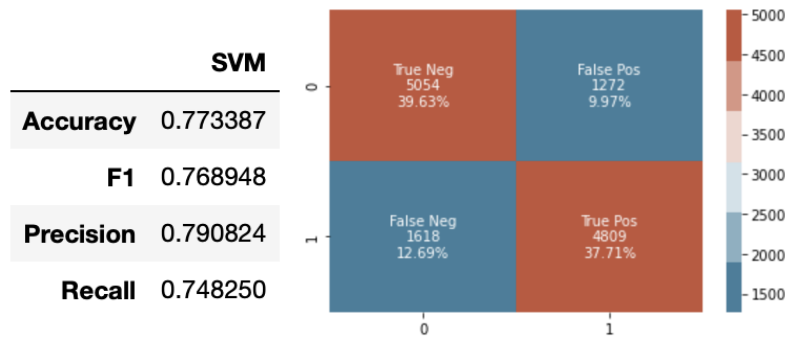
When looking at the results for logistic regression, we see that there is about 50% accuracy, which appears to be the same as random guessing. When looking at the results further, it appears that the model consistently predicted that it would not rain, even when that was not the case. This can be deduced by the precision being 0.5 and the recall value being 1.0. If the model had a higher precision score, this would have meant that the quality of positive predictions would be higher. In order to improve this for the future, some considerations could be taking a closer look at the testing set and ensuring that all of the training data does not fall into the same class, as well as making sure that there are not any errors in the rest of the algorithm.



The Naive Bayes Classifier presented better results than the Logistic Regression. All of the metrics fall above 0.5, so the model appears to be performing better than random guessing as for example, the ratio of correct predictions to total predictions is .74 to 1. The recall in this model is .71, so even though it is predicting less of the true positives, this is balanced out by the fact that the model does not always default to predicting that it will rain tomorrow like for the logistic regression classifier.

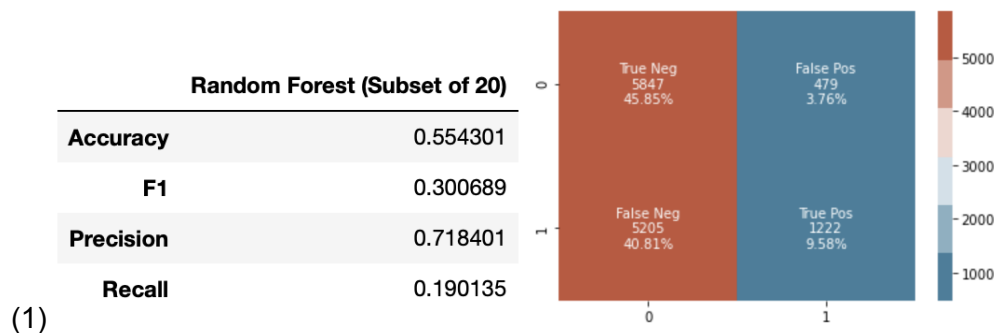


For the SVM regressor, it appears that there is an even higher performance across all metrics when compared to the Naive Bayes. There is 77% accuracy meaning that 77% of the total guesses are accurate out of the total guesses. The precision is also close to 80% meaning that when the classifier says it will rain, it is correct close to 80% of the time. The recall is close to 75% percent which means that the model found ~75% of the points where it did end up raining and identified it as such.

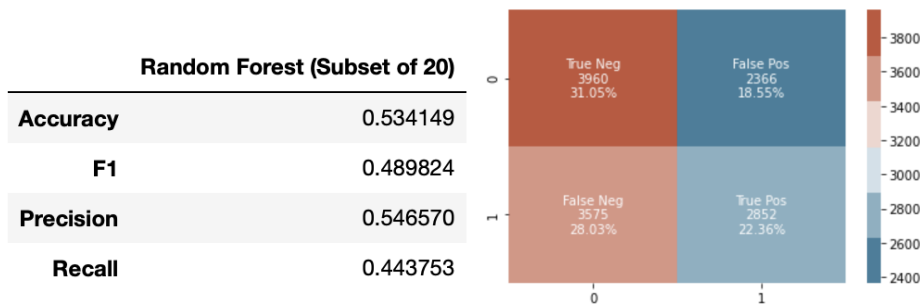


Our RandomForest Model showed some interesting results. Throughout the process, there were struggles with reducing complexity of the model, and that struggle continued as we worked to create a model that ran in a reasonable amount of time. Instead, for this model, we ran a subset of data with 20 entries on a Random forest with 3 trees and max depth of 3 (1), a Random forest with 5 trees and max depth of 3(2), and a Random forest with 5 trees and max depth of 5 (3). This was able to give us reasonable data which we were able to see how RandomForest is able to reduce variability in a number of ways which in turn generally improved our scores, with potentially some drawbacks when we look at model (3) which has begun over-guessing Positive results.

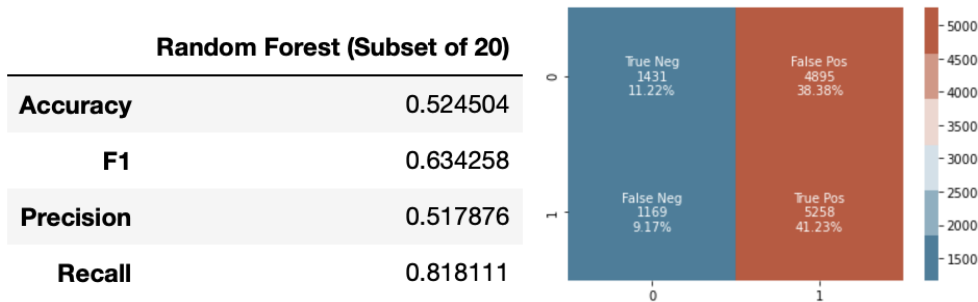
Obviously, this model has much more room for improvement both by the hand coding involved, and also many more interesting explorations that could be made into the effect of different portions of data on the results of our model. Specifically, since we don't know immediately which training data we used for the models (though it is the same set for each model) I'd want to see what imbalances the data shows between the y columns and also in what features were similar and dissimilar.



(2)



(3)



Wrapping up, SVM performed the best overall with Naive Bayes closely behind. The Logistic Regression model needs to be investigated to see if there are metrics to alter to assist with reducing the positive guesses from the model. Random Forest could also use considerable improvement and could potentially be a top scorer if it was continued to be worked with.

It is easy to see that our models made errors- Logistic Regression had random guessing and Random Forest had increased time complexity. Even the Naive Bayes and SVM models weren't perfect. One aspect we wanted to focus on was human error- the models are mathematical, but that doesn't mean they aren't influenced by the efficiency of the coders who write them. For example, the Random Forest follows the correct algorithm, but experiences issues when processing too much data. The other important factor might be the data. Although we balanced the Yes and No results of our data, there is a big discrepancy between different occurrences of different features in the dataset. We could attribute some of the models' imperfections to the handlings of data distribution.

Speaking of data, the features we use can also have an effect on the performance of models. For example, measurements such as WindGust are taken multiple times a day and might be less important to the model in the current format. Although the average wind gust and the change from day to night could be important, the way the data is presented may require more complex calculations to arrive at the same conclusions. This is why we found the less linked and correlated data such as temperature and pressure to provide more importance to the model as-is.

V. Conclusion

This project has taught us many lessons about the ways that we build models, and the assumptions we work with when it comes to choosing both pretrained models and those we

write ourselves. It was very challenging to write the models by hand- it required utilization of mathematical concepts and understanding how our transformations of the data created a path or algorithm a computer could use to 'find' the answer, or in this case class, to our problem. We also had to take time to understand how to communicate data and insights in a way each other and graders could understand it. This was a valuable way to interact with the class and its goals, but it's an understanding that can extend past the timeline of the project.

As far as future work is concerned, our work on this project taught us a lot about the creation of models and looking at the data we work with before using it. Given a longer deadline, our group wishes to delve further into the ways correlation affects data. It was obvious a lot of different factors, such as the wind changes throughout the day, were often correlated with each other. But, what if we instead recorded the positive and negative change maybe throughout the day for wind gusts? Taking a piece by piece approach to this data could give plenty of individual insight into the small-scale particulars this weather data covers and might be able to make us better at handling data and models in the future.

VI. Team Member Contribution

Abby:

- Data Cleaning, Data Exploration
- Naive Bayes and Random Forest Implementation
- Slide Show recording

Rida:

- Data Cleaning, Data Exploration
- Logistic Regression Classifier & SVM regressor
- Slideshow creation